

Class 09: Structural Bioinformatics (part 1)

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The PDB Database

The main repository for biomolecular structure data is the Protein Data Bank (PDB)
<http://www.rcsb.org>

Let's have a quick look at the composition of this database:

```
stats <- read.csv("Data_Export_Summary.csv")
stats
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	176,378	20,438	12,709	342	221
2	Protein/Oligosaccharide	10,284	3,396	34	8	11
3	Protein/NA	9,007	5,931	287	24	7
4	Nucleic acid (only)	3,077	200	1,554	2	15
5	Other	174	13	33	3	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron	Other	Total			
1	83	32	210,203			
2	1	0	13,734			
3	0	0	15,256			
4	3	1	4,852			
5	0	0	223			
6	0	4	22			

```
stats$X.ray
```

```
[1] "176,378" "10,284" "9,007" "3,077" "174" "11"
```

```
as.numeric(sub(",", "", stats$X.ray))
```

```
[1] 176378 10284 9007 3077 174 11
```

This is annoying lets try a different import function from the **readr** package.

```
library(readr)
```

Warning: package 'readr' was built under R version 4.3.3

```
stats <- read_csv("Data_Export_Summary.csv")
```

Rows: 6 Columns: 9

-- Column specification -----

Delimiter: ","

chr (1): Molecular Type

dbl (4): Integrative, Multiple methods, Neutron, Other

num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
stats
```

A tibble: 6 x 9

	`Molecular Type` <chr>	`X-ray` <dbl>	EM <dbl>	NMR <dbl>	Integrative <dbl>	`Multiple methods` <dbl>	Neutron <dbl>
1	Protein (only)	176378	20438	12709	342	221	83
2	Protein/Oligosacch~	10284	3396	34	8	11	1
3	Protein/NA	9007	5931	287	24	7	0
4	Nucleic acid (only)	3077	200	1554	2	15	3
5	Other	174	13	33	3	0	0
6	Oligosaccharide (o~	11	0	6	0	1	0

i 2 more variables: Other <dbl>, Total <dbl>

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
n.total <- sum(stats$Total)
n.xray <- sum(stats$`X-ray`)
n.em <- sum(stats$EM)

round(n.xray/n.total * 100, 3)
```

```
[1] 81.432
```

```
round(n.em/n.total * 100, 3)
```

```
[1] 12.271
```

81.432% of the structures in PDB are solved by x-ray and 12.271% of structures are solved by electron microscopy.

Q2: What proportion of structures in the PDB are protein only?

```
round(stats$Total[1]/n.total, 4)
```

```
[1] 0.8605
```

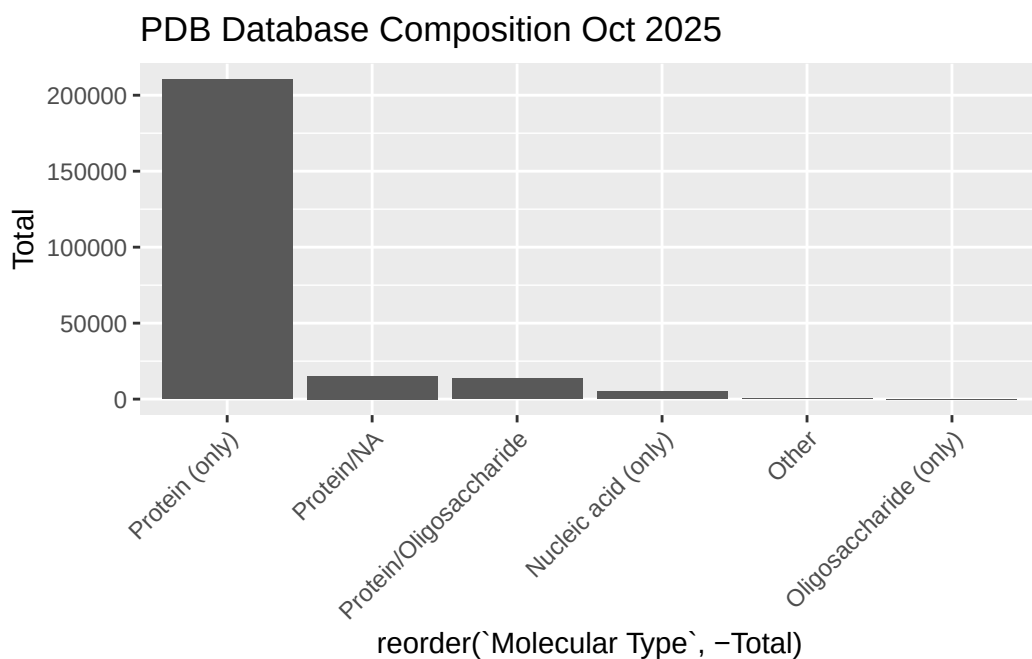
0.8605 of the the structure in PDB are protein only.

Q3. Make a bar plot overview of Molecular type composition using ggplot

```
library(ggplot2)

experiment_method <- colnames(stats)
experiment_method <- experiment_method[-8]

ggplot(stats) +aes(reorder(`Molecular Type`, -Total), Total) +geom_col() +labs(title = "PDB Molecular Type Composition")
```



Visualizing structure data

The Mol* viewer is embedded in many bioinformatics websites. The homepage is <https://molstar.org>

I can insert any figure of image file using markdown format.

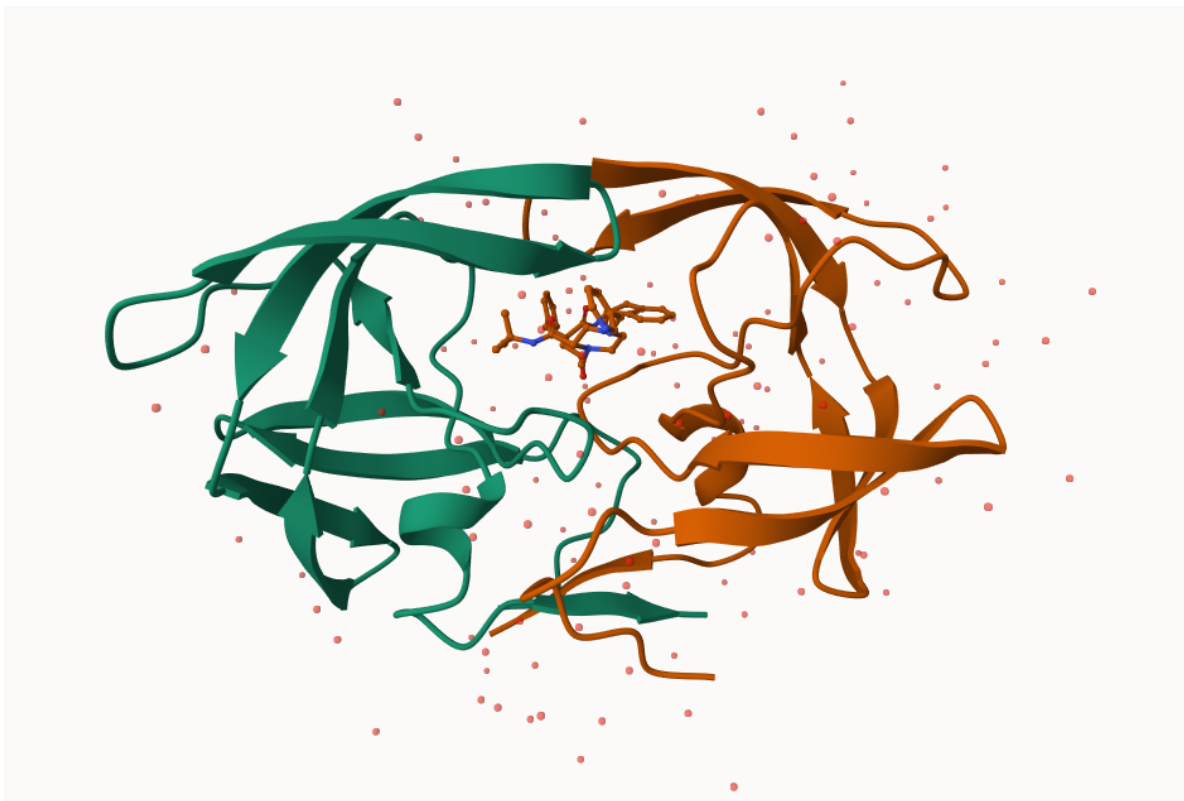
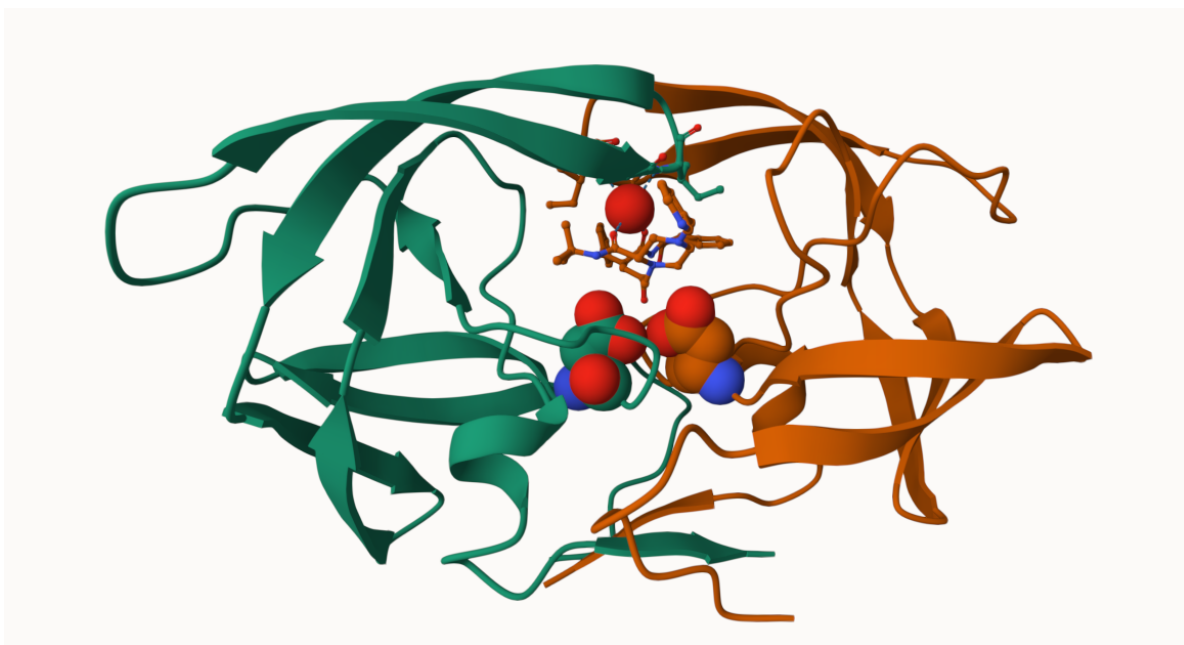


Figure 1: The HIV-Pr dimer with bound inhibitor

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?



There are 2 protease structures in the current PDB (it is a dimer)

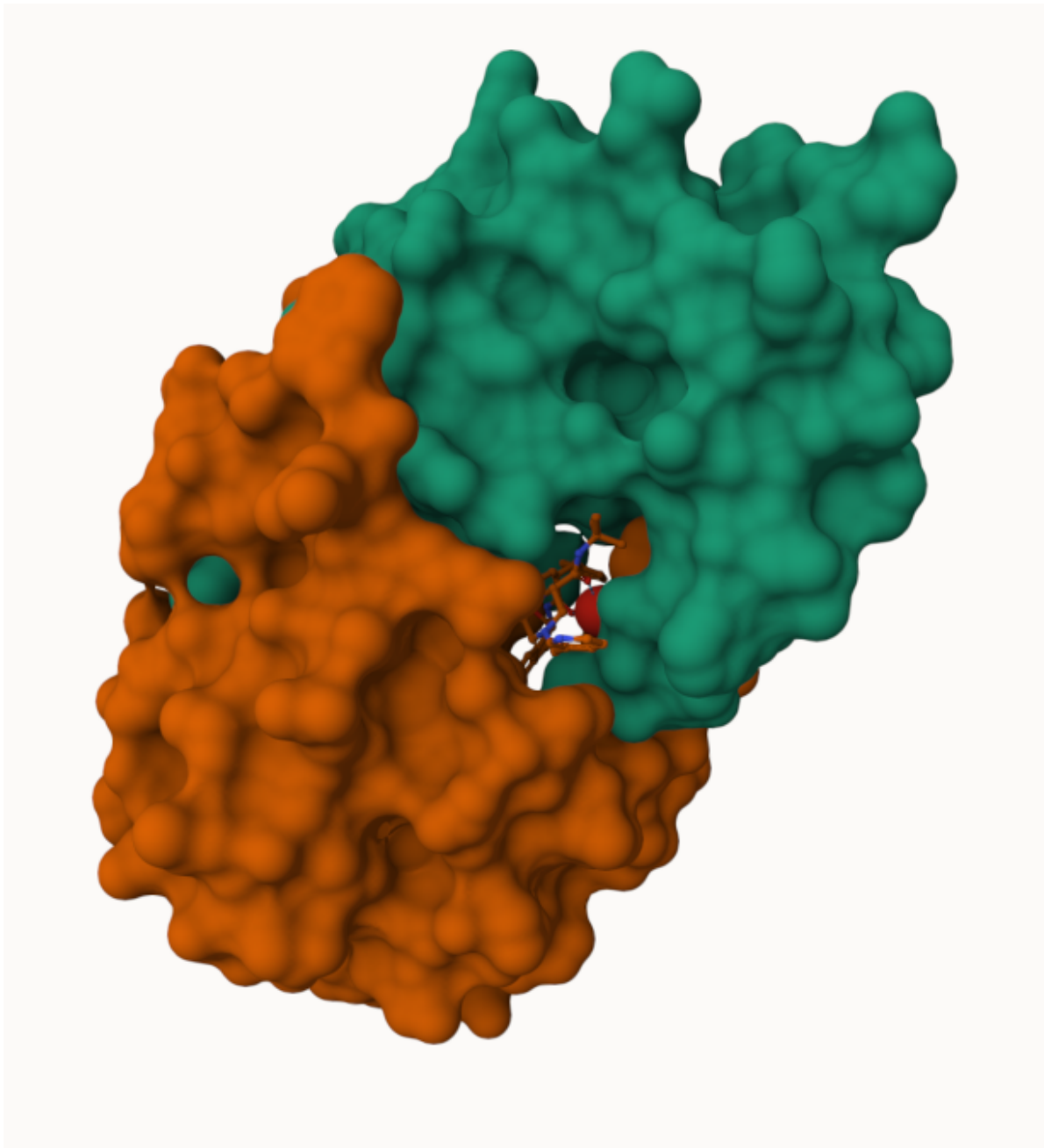
Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

We only see one atom per water molecule because the hydrogens are so small they don't show up when using xray.

Q5: There is a critical "conserved" water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have

The critical conserved water molecule is residue 308.

Q6: Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend "Ball & Stick" for these side-chains). Add this figure to your Quarto document.



Bio3D package for structural bioinformatics

We can use the bio3d package to read and analyze biomolecular data in R:

```
library(bio3d)
```

Warning: package 'bio3d' was built under R version 4.3.3

```
hiv <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
hiv
```

Call: read.pdb(file = "1hsg")

Total Models#: 1

Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)

Protein Atoms#: 1514 (residues/Calpha atoms#: 198)

Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

Non-protein/nucleic Atoms#: 172 (residues: 128)

Non-protein/nucleic resid values: [HOH (127), MK1 (1)]

Protein sequence:

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

There are 198 amino acid residues in this object. Each chain is 99 amino acids long.

Q8: Name one of the two non-protein residues?

MK1 is one of the two non-protein residues

Q9: How many protein chains are in this structure?

There are two protein chains in this structure. It is a dimer.

```
head(hiv$atom)
```

	type	eleno	ety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	esley	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

Let's get the sequence

```
pdbseq(hiv)
```

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
"P"	"Q"	"I"	"T"	"L"	"W"	"Q"	"R"	"P"	"L"	"V"	"T"	"I"	"K"	"I"	"G"	"G"	"Q"	"L"	"K"
21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
"E"	"A"	"L"	"L"	"D"	"T"	"G"	"A"	"D"	"D"	"T"	"V"	"L"	"E"	"E"	"M"	"S"	"L"	"P"	"G"
41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
"R"	"W"	"K"	"P"	"K"	"M"	"I"	"G"	"G"	"I"	"G"	"G"	"F"	"I"	"K"	"V"	"R"	"Q"	"Y"	"D"
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
"Q"	"I"	"L"	"I"	"E"	"I"	"C"	"G"	"H"	"K"	"A"	"I"	"G"	"T"	"V"	"L"	"V"	"G"	"P"	"T"
81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	1
"P"	"V"	"N"	"I"	"I"	"G"	"R"	"N"	"L"	"L"	"T"	"Q"	"I"	"G"	"C"	"T"	"L"	"N"	"F"	"P"
2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
"Q"	"I"	"T"	"L"	"W"	"Q"	"R"	"P"	"L"	"V"	"T"	"I"	"K"	"I"	"G"	"G"	"Q"	"L"	"K"	"E"
22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41
"A"	"L"	"L"	"D"	"T"	"G"	"A"	"D"	"D"	"T"	"V"	"L"	"E"	"E"	"M"	"S"	"L"	"P"	"G"	"R"
42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61
"W"	"K"	"P"	"K"	"M"	"I"	"G"	"G"	"I"	"G"	"G"	"F"	"I"	"K"	"V"	"R"	"Q"	"Y"	"D"	"Q"
62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81
"I"	"L"	"I"	"E"	"I"	"C"	"G"	"H"	"K"	"A"	"I"	"G"	"T"	"V"	"L"	"V"	"G"	"P"	"T"	"P"

```

82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99
"V" "N" "I" "I" "G" "R" "N" "L" "L" "T" "Q" "I" "G" "C" "T" "L" "N" "F"

```

Let's trim to chain A and get just it's sequence:

```

chainA <- trim.pdb(hiv, chain = "A")
chainA.seq <- pdbseq(chainA)

```

Let's blast

```

blast <- blast.pdb(chainA.seq)

```

```

Searching ... please wait (updates every 5 seconds) RID = G5EGTBTU016
.....
Reporting 249 hits

```

```

head(blast$hit.tbl)

```

	queryid	subjectids	identity	alignmentlength	mismatches	gapopens	q.start
1	Query_2951907	1W5V_A	100.00	99	0	0	1
2	Query_2951907	2FDE_A	100.00	99	0	0	1
3	Query_2951907	1AJV_A	100.00	99	0	0	1
4	Query_2951907	2R38_A	98.99	99	1	0	1
5	Query_2951907	2R3T_A	98.99	99	1	0	1
6	Query_2951907	1HXB_A	98.99	99	1	0	1

	q.end	s.start	s.end	eval	bitscore	positives	mlog.eval	pdb.id	acc
1	99	12	110	1.38e-67	199	100	153.9511	1W5V_A	1W5V_A
2	99	2	100	1.70e-67	198	100	153.7426	2FDE_A	2FDE_A
3	99	1	99	1.99e-67	198	100	153.5851	1AJV_A	1AJV_A
4	99	1	99	2.50e-67	198	100	153.3569	2R38_A	2R38_A
5	99	1	99	2.50e-67	198	100	153.3569	2R3T_A	2R3T_A
6	99	1	99	2.50e-67	198	100	153.3569	1HXB_A	1HXB_A

Plot a quick overview of blast results

```

hits <- plot(blast)

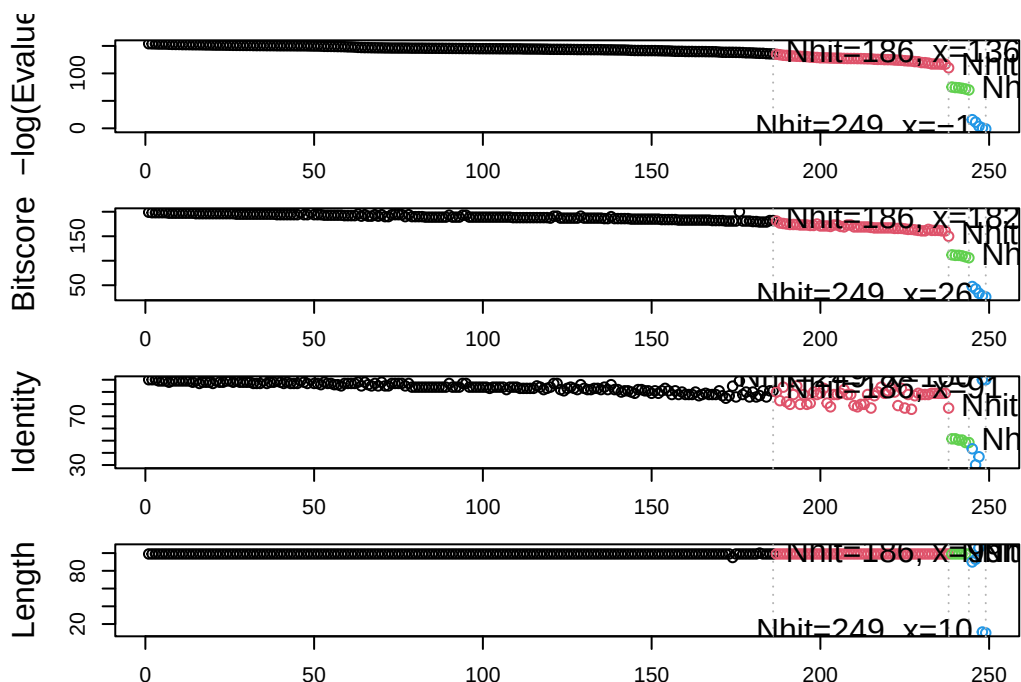
```

```

* Possible cutoff values:    135 110 69 -2
    Yielding Nhits:         186 238 244 249

* Chosen cutoff value of:    69
    Yielding Nhits:         244

```



hits\$ pdb.id

```
[1] "1W5V_A" "2FDE_A" "1AJV_A" "2R38_A" "2R3T_A" "1HXB_A" "1BV9_A" "1AAQ_A"
[9] "1AXA_A" "1HVS_A" "1ZP8_A" "2QHC_A" "1A8G_A" "204L_A" "5COK_A" "1TCX_A"
[17] "2Z54_A" "1D4S_A" "1BV7_A" "1BWA_A" "1A9M_A" "2FLE_A" "1ODY_A" "1GNN_A"
[25] "1GNM_A" "5YRS_B" "1HEF_E" "1ODX_A" "4QGI_A" "1BVE_A" "2AZ8_A" "1A30_A"
[33] "6DH6_A" "6DH0_A" "2I4D_A" "60OS_A" "1RL8_A" "5YRS_A" "1ZSF_A" "2Q64_A"
[41] "6DH3_A" "2NPH_A" "2Q63_A" "1LZQ_A" "1FB7_A" "1G6L_A" "1HIV_A" "600U_A"
[49] "1HVC_A" "2I4V_A" "2AZ9_A" "60OT_A" "2P3B_B" "5KAO_A" "2WLO_A" "6OPT_A"
[57] "1IZI_A" "1MRX_A" "2PYM_A" "2PYN_A" "1DMP_A" "4K4P_A" "1LV1_A" "1AID_A"
[65] "1LV1_A" "1ZBG_A" "3TKG_A" "1HVC_A" "5YOK_A" "1G6L_A" "1FGC_C" "3K4V_A"
[73] "3KT5_A" "3KT5_A" "4QLH_A" "4QLH_A" "2F3K_A" "4Q5M_A" "2AOC_A" "3B80_A"
[81] "3VF5_A" "2AVQ_A" "1DW6_C" "1KZK_A" "2HS1_A" "1K6C_A" "1MTB_A" "4Q1X_A"
[89] "4Q1W_A" "4Q5M_A" "3D1X_A" "2AVM_A" "3PWM_A" "3KT2_A" "3KT2_A" "1SDV_A"
[97] "3JVV_A" "3OY4_A" "1A94_A" "2HS2_A" "4EJ8_A" "2FGU_A" "2AVV_A" "3JW2_A"
[105] "3BVA_A" "1FFF_C" "3S43_B" "2NXD_A" "1FG6_C" "1EBK_C" "4Q1Y_A" "3EL4_A"
[113] "1F7A_A" "1K2B_A" "2FGV_A" "1Z8C_A" "2G69_A" "3EL9_A" "30XV_A" "1BDR_A"
[121] "3N3I_A" "3N3I_A" "30XW_A" "3S43_A" "3EM3_A" "3CYW_A" "5KQX_A" "2B60_A"
[129] "7DOZ_A" "1K2C_A" "1MT7_A" "3EM4_A" "4QJ9_A" "1BDL_A" "3LZS_A" "5T84_A"
[137] "4DQB_A" "7DOZ_A" "4QJ2_A" "3LZV_A" "1SGU_A" "2FXE_A" "1BDQ_A" "3U71_A"
[145] "2R5P_A" "40BD_A" "7MAS_A" "3IXO_A" "3D3T_A" "5YOJ_A" "3LZU_A" "4NJS_A"
[153] "3EKP_A" "1B6J_A" "3EKQ_A" "2RKF_A" "1C6X_A" "7MAR_A" "4DQF_A" "1RPI_A"
```

```

[161] "30U1_B" "3PJ6_A" "2P3A_A" "60GQ_A" "30Q7_A" "5KR1_A" "30QD_A" "4RVI_A"
[169] "30QA_A" "1B6K_A" "30UD_B" "6MK9_A" "3S09_A" "1Q9P_A" "6I45_A" "7SEP_A"
[177] "4NJT_A" "3BXR_A" "4YOA_A" "4DQC_A" "2FDD_A" "2RKG_A" "4DQH_A" "2P3C_A"
[185] "4EP2_A" "4EP2_A" "4EQO_A" "4NPT_A" "60PU_A" "4NPU_A" "3U7S_A" "3HAW_A"
[193] "2AZB_A" "3TTP_A" "3HBO_A" "3GGU_A" "7N6T_A" "60PV_A" "4EQO_A" "60PX_A"
[201] "204N_A" "5T2E_A" "3UCB_A" "3KA2_A" "3FSM_A" "60PW_A" "2AZC_A" "3FSM_A"
[209] "3HLO_A" "2P3D_A" "3T3C_A" "7MYP_A" "6054_X" "60PY_A" "4Z4X_A" "60PZ_A"
[217] "2JE4_A" "1DAZ_C" "7MAP_A" "7MAQ_A" "1K1U_A" "2B7Z_A" "3MWS_A" "1K1T_A"
[225] "8DCH_A" "3I2L_A" "6P9A_A" "2FXD_A" "2J9J_A" "3DCK_A" "2J9J_B" "3NXE_A"
[233] "2040_A" "2040_A" "3NXE_A" "3KA2_A" "3HLO_A" "5B18_A" "1SIP_A" "2SAM_A"
[241] "1AZ5_A" "1SIV_A" "1HII_A" "1IVP_A"

```

Prediction of functional motions

We can run a Normal Mode Analysis (NMA) to predict large scale motions/flexibility/dynamics of any biomolecule that we can read into R.

Q10. Which of the packages above is found only on BioConductor and not CRAN?

the msa package is found in Bioconductor and not CRAN.

Q11. Which of the above packages is not found on BioConductor or CRAN?

The Grantlab/bio3d-view package is not found in Bioconductor nor CRAN.

Q12. True or False? Functions from the devtools package can be used to install packages from GitHub and BitBucket?

True, functions from the devtools package can be used to install packages from GitHub and BitBucket.

Let's look at ADK

```
adk <- read.pdb("1ake")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

```
adk_A <- trim.pdb(adk, chain = "A")
adk_A
```

```
Call: trim.pdb(pdb = adk, chain = "A")
```

```
Total Models#: 1
```

```
Total Atoms#: 1954, XYZs#: 5862 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1656 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 298 (residues: 242)
```

```
Non-protein/nucleic resid values: [ AP5 (1), HOH (241) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV  
TDELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDR  
IVGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQM  
TAPLIGYYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, helix, sheet, seqres, xyz,  
      calpha, call
```

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

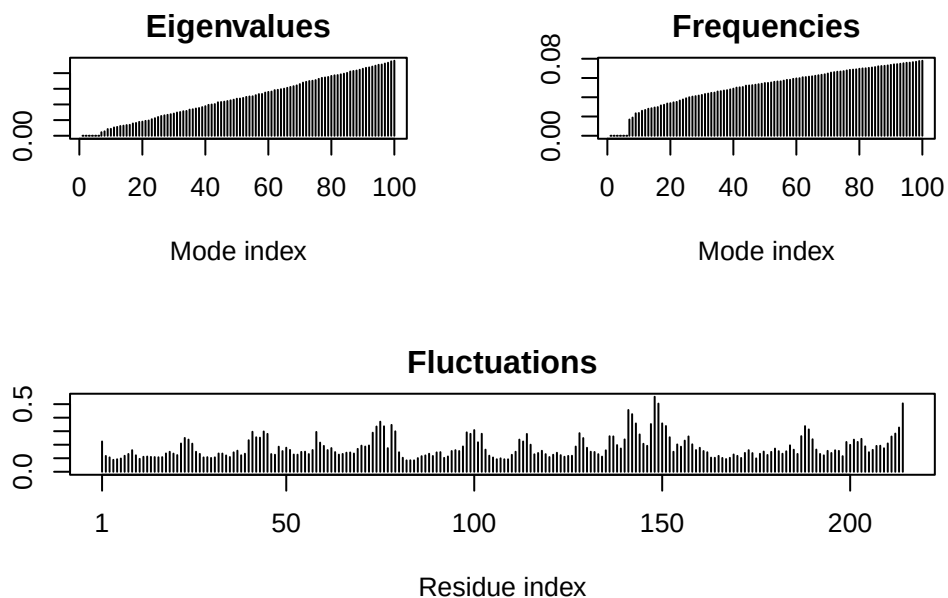
There are 214 amino acids in this sequence.

```
m <- nma(adk_A)
```

```
Building Hessian... Done in 0.08 seconds.
```

```
Diagonalizing Hessian... Done in 0.63 seconds.
```

```
plot(m)
```



Let's write the "trajectory" of predicted motion

```
mktrj(m, file = "adk_nma.pdb")
```

Play with 3D viewing in R

We can use the new **bio3dview** package, which is not yet on CRAN, to render interactive 3D views in R and HTML quarto output reports.

To install from GitHub we can use the **pak** package.

```
#library(bio3dview)
#view.pdb(adk)
```